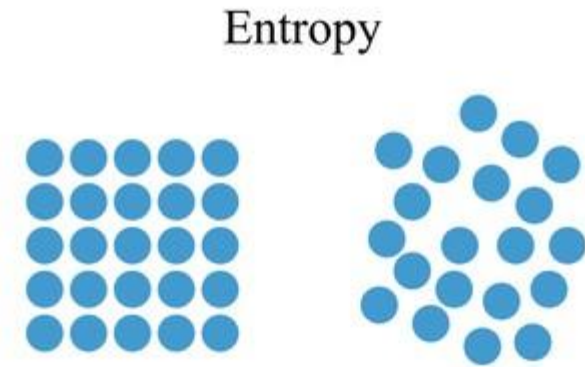


In Machine Learning, **entropy measures the amount of impurity, chaos, or uncertainty** in a dataset.

It tells you how mixed up your data classes are. A dataset with only one class has **zero entropy** (pure), while an equal mix of classes has **maximum entropy** (impure).



Class 50
students

```
graph TD; A[Class 50 students] --> B[10 yes]; A --> C[40 No]
```

10 yes

40 No

Entropy : it defines Randomness and disorders

$$Entropy = - \sum_{i=1}^N p_i \log_2 p_i \quad (9y, 5n)$$

Entropy of Root Node(9y , 5n)

$$Entropy = (-P_y) * (\log_2(P_y)) + (-P_n) * (\log_2(P_n))$$

$$= \left(- \left(\frac{9}{14} \right) \right) * \left(\log_2 \left(\frac{9}{14} \right) \right) + \left(- \left(\frac{5}{14} \right) \right) * \left(\log_2 \left(\frac{5}{14} \right) \right)$$

$$= (-0.64) * (\log_2(0.64)) + (-0.35) * (\log_2(0.35))$$

$$= (-0.64) * (-0.64) + (-0.35) * (-1.49)$$

$$Entropy = 0.94$$

Entropy Scale (For Binary Classification)

Entropy = 0 (Pure): All data points belong to a single class (e.g., 14y, 0n). The outcome is 100% predictable.

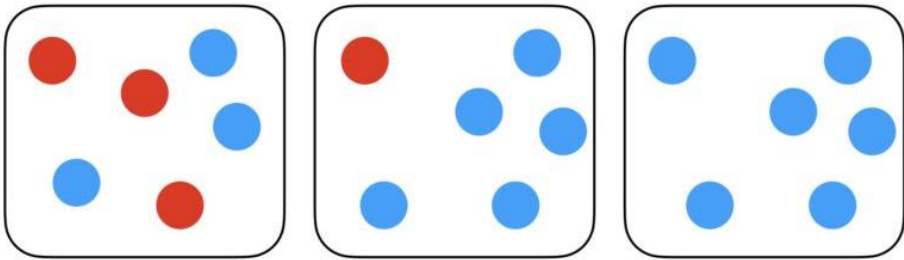
Entropy = 1 (Impure): Data points are split evenly between classes (e.g., 7.5y, 7.5n). Uncertainty is at its highest.

0 < Entropy < 1: A partial mixture (e.g., 9y, 5n gives \$0.94\$, 2y, 12n gives \$0.59\$).

Gini Impurity is a measure used in decision trees to determine how impure or mixed a dataset is. It represents the probability of incorrectly classifying a randomly chosen sample if it is labeled according to the class distribution in the dataset.

$$Gini = 1 - \sum_{i=1}^n (p_i)^2 \quad (9y, 5n)$$

$$Gini = 1 - \left[\left(\frac{y}{Total} \right)^2 + \left(\frac{n}{Total} \right)^2 \right]$$



GI of Root Node(9y , 5n)

$$Gini = 1 - \left[\left(\frac{9}{14} \right)^2 + \left(\frac{5}{14} \right)^2 \right]$$

$$= 1 - [(0.64)^2 + (0.36)^2]$$

$$= 1 - [0.41 + 0.13]$$

$$\mathbf{Gini = 0.46}$$